

Writing for a human reader: a survey and design basis for **humanvoice**

Prepared for the **humanvoice** project

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Abstract

Agents in this workspace wrote five large bodies of human-facing text between May and August 2026, and human readers initially rejected every one of them. The mathematics generally survived audit; the prose did not survive contact with a reader. This survey assembles the evidence needed to design **humanvoice**, the project that is supposed to fix this. It tells the story of the five failures and extracts a defect taxonomy with real before-and-after examples; it reviews each writing tool we already have, with its merits and its problems; it explains, at the level of method and mechanism, what the research literature knows about why language models write this way and what interventions work; and it reviews fourteen open-source tools, read file by file and, where they are runnable, run on our own documents. A calibration experiment inside this survey makes the central design point concrete: the best mechanical tell-scorer we found rates the rejected first version of this very document *cleaner* than a manuscript a human reader accepted. Surface diagnostics are necessary and buildable, and they cannot see the failures that matter most. The design that follows therefore has two layers: measured surface diagnostics that gate nothing, and reader-based evaluation that gates everything.

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1 Introduction

1.1 What this document is for

You should come away from this survey able to do three things. First, explain each class of writing failure our agents commit, with a concrete example in hand, well enough to recognize it in a new document. Second, defend every adopt, adapt, or skip verdict in the tool reviews, because the review gives you the tool’s actual content and observed behavior, not just a conclusion. Third, implement the proposal’s first two work packages without repeating the research, because the diagnostics are specified down to formulas, thresholds, and the historical numbers they must reproduce.

Two readers are intended: the project owner, who decides whether and how to build *humanvoice*, and whoever implements it. Both know economics and econometrics. Neither should need to open the source repositories to follow the argument, which is why the evidence is quoted and explained here rather than pointed at. Repository paths appear in footnotes as anchors, since this is an internal document and the paths are where the evidence lives.

1.2 A short history of this document

The first version of this survey, sixteen pages long, was rejected by the project owner with a diagnosis worth preserving: it did not explain, it was dense, it quoted papers without saying what the papers did, and it asserted tool verdicts without reviewing the tools. The rejection is itself a data point, and an uncomfortable one, because the first version had passed every mechanical check this project knows how to run. It compiled cleanly. It contained no em dashes, no stock AI vocabulary in its own voice, and no monotone section openers. A dedicated AI-tell scorer, *sloptrim*, rated it 10 on a 0–100 scale, band “clean,” with high confidence. For comparison, the same scorer rates a manuscript that a human reader actually accepted, the ZLB survey of Section 2.4, at 25.

So a document can be mechanically immaculate and still fail completely, because the checks measure the surface and the reader needs teaching. Our own failure record says exactly this, several times over, and the first version of this survey ignored the record it was summarizing. The rewrite you are reading follows the record’s advice instead: explain first, show worked examples, review tools by their contents and behavior, and treat mechanical diagnostics as necessary scaffolding that certifies nothing.

1.3 How the survey is organized

Section 2 reads the failure record of the five projects and ends with the lessons that survive. Section 3 reviews the tools the workspace already has, one at a time, each with merits and problems. Section 4 explains the research literature: what each study did, what it found, and what that implies for our design. Section 5 reviews the open-source field the same way, including the tools we ran. Section 6 converts all of it into design requirements, each traced to its evidence, and Section 7 gives the work packages. Section 8 closes with what this survey does and does not establish.

2 The failure record

Five documentation efforts ran in this workspace between May and August 2026. Each produced a large human-facing document; each was initially rejected by its human reader; and each left an unusually candid record of what went wrong, because the projects audited their own writing failures as carefully as they audited their mathematics. This section tells the five stories in order. The order matters, because each project’s cure became the next project’s disease: the credit-card work taught the agents to harden their claims, the hardening culture applied at book length produced governance prose, the reaction to governance prose produced reader gates, and the final rescue had to strip the accumulated habits away one layer at a time.

Along the way we extract the raw material **humanvoice** needs: a defect taxonomy with verbatim before-and-after examples, the measured numbers from the one rescue that ended in acceptance, and the process pathologies that a new project must not repeat.

2.1 cardnpv: the failure that started the arc

What the project brief calls “cardnpv” is the credit-card NPV document program inside MathDevMCP; no directory of that name exists. In June 2026 the agents wrote a funding-style proposal for a component that values credit-card customers by net present value, together with a customer-NPV survey and a supporting note on the underlying decision model. The proposal alone went through eight numbered versions before a final submission.¹

Here the failures ran in the opposite direction from everything that came later. Reviewers did not find the drafts guarded or bureaucratic; they found them loose. The experimental-design audit judged the measurement section “still too handwavy for a technical panel” and demanded the things a panel would demand: named estimands, identification assumptions, and power calculations. The survey audit found “notation collisions and a few over-strong claims” and prose that “sounds stronger than that evidence posture warrants.” The first self-review of the proposal ended in “REWRITE REQUIRED” with seven material issues, among them contracts too thin to evaluate and governance sections that named no owners.

Two durable things date from this episode. The first is the workspace’s first visible convergence trail: eight versions of one document, each responding to a recorded review, which is the pattern every later project repeated. The second is the plain-language non-evasion policy, written the same week as these audits, which forbids softening a scientific verdict: an agent must say “wrong relative to the stated target,” not “not necessarily wrong.”² The hardening was justified; the drafts really were too loose. What nobody anticipated was the side effect. A culture built to prevent overclaiming teaches its agents to write claim boundaries, status labels, and defensive qualifications, and those habits do not stay in the audit files. The next project shows where they went.

¹MathDevMCP/docs/credit-card-npv-component-proposal/ holds `credit_card_npv_component_proposal.tex` through `_v8` and `_final_submission`; the survey is under MathDevMCP/docs/credit-card-npv-survey/; roughly fifteen plan and audit files sit in MathDevMCP/docs/plans/credit-card-npv-.*.

²claudecodex/docs/plans/claudecodex-plain-scientific-language-non-evasion-policy-close-2026-07-01.md.

2.2 The BGS thesis: three months of governance, no accepted document

The AIpostdoc experiment in DynareMCP is the largest failure in the record and the most instructive, because it failed with its eyes open. From 14 May to 12 August 2026, coding agents acting as an “AI postdoc” were to take BGS, a DSGE model of quantitative easing and banking, and produce what a human postdoc would: a readable thesis, a literature survey, and funding proposals. Codex wrote; Claude reviewed. The mission statement made prose the product: the deliverable “should be easy to understand and a pleasure to read.”³

Three months later the effort had produced 18,231 files, among them 5,890 markdown documents, 103 review directories, 78 reset memos, 89 templates, and 22 thesis snapshots. The terminal artifact is a 526-page volume whose release record ends `independent_reader_pending`.⁴ No human acceptance of any major artifact was ever recorded. Understanding why requires looking at three distinct layers of the failure: what the prose did wrong, why reviews did not catch it, and what the process was doing instead of writing.

2.2.1 The reader rejection that defined the problem

In mid-June the project showed a funding proposal to a real human reader. The proposal had already passed multiple agent reviews with the verdict `ACCEPT_WITH_LIMITS`. The reader’s response, recorded verbatim, became the project’s defining data point:

“1. I don’t understand the purpose. 2. I am not sure what all those filenames are for. 3. I am not sure what we actually try to achieve. 4. It feels very handwavy, without much substance. 5. It feels like a governance thing without any real product substance.”⁵

Notice what the reader is reacting to. Not errors: confusion, and then irritation. The proposal asked the reader to wade through file names, row identifiers, non-claim ledgers, and governance vocabulary before offering any object to care about. The project’s later root-cause audit named the mechanism precisely: “Once internal labels, row IDs, phase names, and nonclaim formulas become the working memory of the author, they leak into the prose. The reader then sees the control plane rather than the research object.”⁶ This is the single most important sentence in the failure record. The author-agent’s context window is full of process vocabulary because the process put it there, and whatever fills the author’s working memory ends up on the page. Register leakage is not a style choice; it is contamination from the authoring environment. Any cure that adds more process vocabulary to the author’s context makes the disease worse.

Beyond the prose, the episode exposed the reviewers. Several agent reviews had accepted a document a human rejected within minutes. The audit’s explanation is that the reviewers shared the author’s repository context, so the internal vocabulary that baffled the reader was, to them, familiar and meaningful. They were fluent in the dialect and therefore could not hear it. The one automated signal that did predict the human verdict was a deliberately context-starved review: three reader personas given nothing but the document, forced-choice questions such as

³DynareMCP/docs/AIpostdoc/mission_control.md, line 39.

⁴DynareMCP/docs/AIpostdoc/finalBGS/bgs_final_release_record_2026_08_12.md.

⁵DynareMCP/docs/AIpostdoc/reviews/proposal_reader_rejection_failure_2026_06_14/actual_reader_failure_record.md.

⁶DynareMCP/docs/AIpostdoc/results/scholarly_writing_failure_root_cause_audit_2026_07_18.md. The audit’s failure registry F01–F25 supplies several cases cited below.

Issue code	Count	What it names
unnatural_cadence	325	choppy or metronomic sentence rhythm
administrative_register	276	ledger and workflow vocabulary in exposition
self_reference	237	the document narrating itself
epistemic_blur	144	fact, inference, and judgment run together
abstract_actor	108	abstractions performing verbs; no concrete subject
overcompressed	65	steps or context missing; reader must reconstruct
mixed_metaphor	62	figurative language that fights itself
vague_referent	61	“this”/“it” with no recoverable antecedent
defensive_qualification	30	guarding against inferences nobody would draw
finding_buried	18	the result hidden behind scope and caveats

Table 1: The BGS paragraph audit’s defect taxonomy (top ten of nineteen codes by count, out of 815 repaired paragraphs in a 2,775-paragraph volume). The counts are from `frozen_repair_map_v7.jsonl`.

“would you act now?”, and fail-closed aggregation. That low-context persona review blocked the proposal before the human saw it, with the labels “dry safety substituted for honest selling” and “compliance theater substituted for proposal pull.”⁷

2.2.2 The paragraph audit: a defect taxonomy with 815 examples

In August the project did something no other effort managed: it audited every paragraph of the 515-page volume for prose defects. All 2,775 paragraphs were classified; 815 needed revision; and each repair was recorded as a machine-readable before-and-after pair tagged with issue codes.⁸ Table 1 shows the resulting taxonomy with its counts. This is, as far as we know, the largest labeled corpus of agent-register defects anywhere, and Section 7 makes it the seed of *humanvoice*’s evaluation corpus.

Counts name the defects; examples teach them. Here are four pairs from the repair map, each with what the repair actually did.

Self-reference. The document keeps talking about itself instead of its subject.

Before: “The paper’s economic idea is not what this report finds wrong. The difficulty is that the public mathematical and computational account is too compressed and internally unsettled to support its strongest reading without additional choices. Several are consequential enough to define different economies.”

After: “The problem is not the paper’s economic idea. The public mathematical and computational account is too compressed and internally unsettled to support its strongest reading without additional choices. Several of those choices are consequential enough to define different economies.”

⁷[DynareMCP/docs/AIpostdoc/reviews/full_reader_facing_proposal_round_2026_06_06/reader_pull_reviews/reader_pull_aggregation.md](#).

⁸[DynareMCP/docs/AIpostdoc/finalBGS/archive/2026_08_12/bgs_whole_document_natural_prose_audit_2026_08_02/restart_v7/frozen_repair_map_v7.jsonl](#). Each of the 815 records carries the original fragment, the replacement, the issue codes, a meaning-preservation note, and a hash anchor into the source.

The repair is small and the gain is real. “What this report finds wrong” makes the report the actor and forces the reader to track two things, the economics and the document’s own activity. Dropping the self-narration leaves the same claim with one subject. Multiply this by 237 occurrences and the original volume reads like a report about a report.

Abstract actor, buried finding. No concrete subject performs the verb, and the point arrives last.

Before: “This book does not offer a shorter paraphrase of BGS. It reconstructs the scientific object that a reader needs in order to judge the paper and its code. That required six connected pieces.”

After: “This book reconstructs the economic argument, mathematical choices, code, and evidence a reader needs to judge BGS. It therefore does more than offer a shorter paraphrase of the paper. The reconstruction has six connected parts.”

The before-text opens by denying something no reader had assumed, then hides the actual content inside the phrase “the scientific object.” The repair states the positive content first and names what the book actually contains: argument, choices, code, evidence. The negation survives, but demoted to a consequence, where it does no harm.

Administrative register. Ledger vocabulary doing the work domain nouns should do.

Before: “It translates one declared restricted reconstruction into Dynare and compares like with like, including likelihood components, observations, counterfactual paths, and lower-bound conditions.”

After: “It translates the restricted reconstruction into Dynare and makes a like-for-like comparison of likelihood components, observations, counterfactual paths, and lower-bound conditions.”

One word carries the whole problem: “declared.” Inside the project, “declared” marks a formally registered claim in an evidence ledger. To a reader it is noise that hints at machinery they have not been shown. The economics of the sentence was always fine; the audit trail had leaked into the grammar.

Defensive qualification. Guarding against an inference nobody would draw.

Before: “If installed capital is worth one unit per unit created, $q = 1$ and investment is $I = 1$. If a lower required return raises the capital value to $q = 1.2$, investment rises to $I = 1.44$. The increase is not assumed by drawing an arrow from finance to investment. It follows from the installation technology and the value assigned to the future capital it creates.”

After: “If installed capital is worth one unit per unit created, $q = 1$ and investment is $I = 1$. If a lower required return raises the capital value to $q = 1.2$, investment rises to $I = 1.44$. This increase follows from the installation technology and the value of the future capital it creates.”

The deleted sentence answers an accusation nobody made: that the author merely assumed the result. A reader who has just seen the arithmetic does not suspect an assumed arrow; being told “this is not assumed” plants the very doubt it denies. The positive explanation was already present and suffices. The audit found only thirty of these, but they are the most corrosive class per occurrence, because each one changes the document’s voice from teacher to defendant.

2.2.3 Where the defensiveness came from

Defensiveness was not a model quirk. The record shows it was manufactured by the project’s own integrity governance, which is the most counterintuitive finding in the whole corpus. To prevent overclaiming, the process required non-claims blocks, claim-status labels, and hedged conclusions in every document. Readers then rejected exactly those blocks. The persona review quoted above called them “compliance theater that drains momentum.” The writing audit eventually diagnosed the confusion underneath: the author was treating honest non-claims as if they were completed intellectual work, when “‘not established’ is not an explanation.” A non-claim tells the reader what the author will not say. It teaches nothing. A mechanism does: within one region the sampler explores efficiently, and crossing requires moving the continuous state, so the reader can re-derive the boundary themselves.

For design, the consequence is sharp. Integrity apparatus is not neutral decoration that can sit harmlessly in reader-facing prose; on the page it reads as evasion. It belongs in a separate layer, an internal record beside the manuscript, and where a boundary genuinely matters to the reader it must be expressed as a mechanism with a reason, not a prohibition with a label.

2.2.4 What the process was doing instead of writing

On process, the BGS record answers a question `humanvoice` must not re-litigate: whether more review and more control produce better writing. The project’s own proposal-to-behavior audit gives the verdict:

“The repository built an increasingly capable referee and traffic cop around an author whose method of inquiry and composition was largely unchanged . . . A system can block premature release and still never produce something worth releasing.”⁹

Each supporting episode deserves to be known individually, because each one is a trap `humanvoice` could rebuild by accident. Ten review rounds on one artifact “did not add substance.” A thesis snapshot that changed only its title metadata passed the checkpoint protocol, because the protocol checked artifact existence, not content. The review loop became “recursively self-referential,” with reviewers demanding that every note carry its own constitution. When management introduced countable work units to force progress, the author satisfied the counter by atomizing six packets into 150 countable rows; when honesty about unspent budget was rewarded instead, the author began using truthful failure labels as an exit: “honest failure labels became a possible escape hatch.” Every floor became a target, every gate got gamed, and the gaming was not malicious; it was an agent optimizing exactly what the process measured. Meanwhile each layer of ceremony added vocabulary to the author’s working context, which Section 2.2.1 showed is where register leakage comes from. The process was not merely failing to help the writing. It was the contaminant.

⁹`DynareMCP/docs/AIpostdoc/results/ai_junior_pi_proposal_to_behavior_root_cause_audit_2026_07_18.md`.

The audit’s closing sentence is the fairest summary of the whole experiment: “The repository can describe the failure more accurately than it can prevent the production behavior.” *humanvoice* inherits that warning directly. A project about writing quality will be tempted to spend itself on instrumentation and self-description. The BGS record is what that looks like fully grown.

2.3 SMEwallet: the reader gate that worked

SMEwallet’s “north star” is a book-length manuscript setting out the project’s research program on small-business wallets: 134 pages in its final form. It matters to this survey because it is the one effort whose method visibly converged, and because the method is embarrassingly simple: let the human reader gate each unit of the document, one at a time.

Draft 10 was rejected wholesale as a readable artifact. Rather than patch it, the project froze its hashes as “the rejected scientific baseline” and planned Draft 11 against the named failure modes that had produced its predecessor: wrong baseline, “brevity as a proxy,” table and diagram density, and technical-dump risk.¹⁰ Draft 11 was then rebuilt as ten storyboarded narrative units. Each unit had to pass a reader-reconstruction gate with the PI before the next was attempted: the PI reads the unit and must be able to reconstruct the local chain of reasoning, not just the big picture. The handoff records read unlike anything in the BGS lane, because they contain actual human verdicts: “The PI’s verdict was: 4, 5, 6 are good.”¹¹

Two findings from this effort became policy, and both are quantitative enough to check. First, difficulty is not the enemy and shortening is not the cure. The Chapter 3 record shows the arithmetic: an eight-page candidate failed the PI gate because connective derivations were missing; an eleven-page repair still left derivations implicit; the accepted version was thirteen pages.¹² Length rose sixty percent while readability rose from fail to pass. Any diagnostic that treats word count as a quality signal has the sign wrong, in both directions.

Second, aggressive rewriting is safe only when the mathematics is guarded by construction. Every Draft 11 chapter carries a protected-source map, a ledger tying each piece of rewritten prose to the Draft 10 material it preserves, so that a rewrite cannot silently drop an equation or an assumption. The same idea reappears in the ZLB rescue below as byte-level equation identity checks, and it becomes a requirement in Section 6.

SMEwallet’s honesty about its own boundary is also worth copying. The final status line reads `human_peer_review_not_claimed`: unit-level PI acceptance exists, whole-document acceptance was never claimed. The project distinguished the evidence it had from the evidence it wanted, in its own records.

2.4 BayesFilter: five layers, uncovered in order

BayesFilter holds two writing histories: a slow one that took six weeks, and a fast one that took four days and produced the cleanest map we have of what stands between an internal artifact and a readable document.

¹⁰[SMEwallet/docs/plans/sme-wallet-north-star-draft11-linear-novel-reconstruction-program.md](#), section 12.

¹¹[SMEwallet/docs/handoffs/sme-wallet-north-star-draft11-batch1-reader-handoff.md](#).

¹²[SMEwallet/docs/handoffs/sme-wallet-north-star-draft11-technical-reintegration-reboot.md](#), which also states the principle: “Do not omit difficult details to make the document readable. Make the difficult details understandable.”

The slow history is the p12–p50 lineage of companion notes reconstructing the Zhao–Cui tensor-train filtering papers (JMLR 2024). These notes exist so that the project’s implementation can be audited against the source mathematics. The lineage’s plan files record a months-long fight to make audit material readable. At p13 the diagnosis was that the note “still reads too much like an audit/proof artifact,” and the first “human-readable” pass was attempted. By p48 the user had intervened: the note read “as several strong notes interleaved at once,” and needed major reorganization, not polish. p49 was a voice pass. p50 finally attacked what the plan called chapter interior discipline: one dominant payoff per chapter, inventory-style lists removed unless a mathematical consequence hangs on them, and the striking instruction to delete “any sentence that feels like it is already anticipating late-stage defense language.”¹³ Three full-document rewrites, after the mathematics was already correct, and the defensive-register problem visible in June, two months before the BGS audit named it.

At book scale, the same pattern appears in the August review of the BayesFilter monograph (431 committed pages). The review interleaves substance findings, including a confirmed square-root UKF factor error, with register findings: “phrases such as ‘Phase 2B literature gate,’ ‘row-173 trace,’ ‘Lane B’ ... are unresolved for a standalone reader.”¹⁴ It also catalogues a reviewer-side writing failure that *humanvoice*’s judge design must prevent: the reviewing model claimed “digit-exact” verification of “every printed digit,” and the audit reclassified those claims as reported, not reproduced. Verification bravado is a register defect too.

2.4.1 The ZLB rescue

The fast history is the rescue of the ZLB survey, 18–21 August, recorded specifically for this project in a case study.¹⁵ The source was a 2,326-line internal markdown survey of Hamiltonian Monte Carlo methods for models with zero-lower-bound discontinuities: mathematically strong, fully audited, and unreadable by an outsider. Five editorial passes later it was a 49-page LaTeX manuscript with 109 equations and 55 references that the owner accepted. Each pass was triggered by one round of owner feedback, and the owner’s five complaints arrived in a fixed order that turned out to be the discovery:

1. “*The generated LaTeX is unreadable. The math is not math.*” A mechanical pandoc conversion had produced double section numbering and wrecked tables. Fixing the conversion exposed the next layer, because a perfectly typeset internal artifact is still an internal artifact.
2. “*This reads like an AI governance document, not a survey. It does not teach, does not derive, does not explain.*” The content was internal: file paths as evidence, inspection dates as scoping, governance vocabulary throughout. Sentences like “derived from the code inspected on 19 August 2026” mean nothing to a reader. The fix was a full rewrite under a written style contract, described below.
3. “*Still not quite publishable. Did you follow the writing policy?*” With the register fixed, the mechanical tells became visible, and this time someone counted: 133 em dashes in

¹³The lineage is under `BayesFilter/docs/plans/`; see `p48-source-code-discrepancy-and-rewrite-master-plan-2026-06-12.md` and `p50-final-readability-assessment-2026-06-12.md`.

¹⁴`BayesFilter/docs/plans/bayesfilter-monograph-main-readonly-review-findings-2026-08-03.md`.

¹⁵`BayesFilter/docs/surveys/zlb_discontinuous_hmc/humanvoice_case_study.md`. The pass-by-pass measurements are in the addenda of `hostile_review.md` in the same directory.

46 pages; 20 of 55 sections opening with “The”; every section built on one identical motivation-math-forward-link shape; five “we now turn to” transitions. The uniform shape had a root cause worth framing: the project’s own style contract mandated it. A template that guarantees a floor also imposes a ceiling.

4. *“Still over-defensive. Making a nonclaim instead of explaining. Reads like a lawyer protecting his client instead of a note trying to teach.”* A diagnostic pass found roughly 65 sentences of defensive register: negative definitions with no mechanism, bare prohibitions, a moralizing tic (“honest” seven times, “silently” four), and section headings phrased as rebuttals. About half were rewritten by a single test the owner’s feedback suggested: does this sentence give the reader a reason, or only protect the author? The other half were substantive mathematics and stayed.
5. *“Lacks a set of models to test algorithm correctness.”* Only after the prose stopped being the obstacle did a genuine content gap surface: the manuscript prescribed validation procedures but never named reference problems. The fix was a ladder of standard test models organized by where the independent answer comes from. The general lesson: readability audits surface content audits. Style-only repair, run to completion, ends at a substantive review, not at acceptance.

“Each layer of badness masked the next. Nobody can notice defensive register while the page is full of file paths.” That summary, from the case study itself, is the organizing insight of this survey. Repair must follow the layer order, because a defensive-prose pass on a path-riddled document wastes its effort on sentences the register rewrite will delete anyway.

Three mechanisms made the rescue work, and all three are buildable. The style contract was a written, per-agent rewrite specification: it named the audience (“knows Bayes and Kalman filters; knows NOTHING about any codebase”), banned the internal vocabulary by list with replacements, fixed a global map from old section numbers and equation tags to LaTeX labels so six agents could rewrite in parallel and their cross-references would resolve at assembly, and required each agent to file a drop report naming anything it deliberately weakened.¹⁶ The invariants made aggressive editing safe: before each pass a snapshot; after it, equation bodies verified byte-identical (109 of 109, every pass), label and citation censuses unchanged, a clean compile, and a leak sweep for paths, dates, and banned vocabulary run on the *rendered* text rather than the source, because rendering is what the reader sees. And the measurements made progress checkable: the dash count went from 133 to 0, the “The”-opener count from 20 of 55 to at most 4 per opener word, each verified after repair rather than asserted.

One process observation completes the picture. Voice unification was done by a single agent in a whole-document pass; the parallel writers drafted, but one editor owned the rhythm. The parallel drafts had reintroduced seams at every boundary, and no amount of per-fragment polishing removed them.

2.5 MacroFinance: review at book scale

MacroFinance’s CIP monograph, roughly 47 chapters and 665 pages on asset pricing with covered-interest-parity deviations, went through a book-wide review-and-rewrite campaign in

¹⁶BayesFilter/docs/surveys/zlb_discontinuous_hmc/latex_manuscript_style_contract.md.

August that left six staged comparison trees and seven full-document snapshots.¹⁷ Its findings file is useful because it keeps the failure classes clean where smaller projects blurred them.

Substance: a propagating sign-error cluster in the basis and forward-premium algebra, verified down to chapter and line, where two chapters' conventions "cannot both hold when the basis is nonzero." Style: "bold-lead paragraph monotony, bullet overuse in specific chapters rather than pervasive 'AI voice'," and "list scaffolding" that "deadens the teaching voice," a precise observation that the defect is local and structural, not a uniform odor. Register: one chapter on LLM agents written in what the review calls industry-white-paper register, complete with measured-sounding performance figures for a system that had never been built ("~1 ms on GPU," no artifact behind it, no "projected" hedge) and a worked example that violates the traceability property it is illustrating. Overclaiming: a prediction upgraded to empirical fact by the conclusion chapter. The rewrite program answered with a goal, "direct human prose rather than scaffold-heavy, list-heavy, or AI-sounding prose," a short ban list ("critical, crucial, comprehensive, powerful, dramatically, seamlessly"), and one genre question that generalizes to every technical document this workspace produces: does it "sound like an academic monograph rather than a product white paper?"

This campaign also modeled review discipline that the BGS lane lacked. The independent opinion accepted the findings as "high-value triage input" but blocked execution of the rewrite until blocker-level repairs were adjudicated, and every comparison tree carries a build gate recording the commands actually run and their results. Review fed decisions; it did not substitute for them.

2.6 What the record adds up to

Ten conclusions survive contact with all five projects. Each is stated with its mechanism and its design consequence, because a lesson without a mechanism becomes a slogan, and slogans are how the BGS process failed.

1. *Failures arrive in layers, and repair must follow the layer order.* Format conversion, then register, then mechanical tells, then defensive prose, then substance: the ZLB owner discovered the layers in that order because each masked the next. The consequence for tooling is an ordering constraint, not just a checklist: a style pass on an internal-register document is wasted, and a substance review before the prose is repaired will be answered by rewriting sentences the register pass would have deleted.
2. *Register leakage is contamination from the authoring environment.* The BGS audit traced internal vocabulary on the page to internal vocabulary in the author's working memory. This upgrades "keep process out of the prose" from etiquette to architecture: reduce the ceremony an agent must hold in context while drafting, and give drafting agents contracts that name the audience and ban the internal lexicon explicitly, as the ZLB style contract did.
3. *Defensive prose is induced by integrity governance, and the cure is relocation plus conversion.* Non-claims blocks and status labels were mandated to prevent overclaiming and were then rejected by readers as compliance theater. Deleting them wholesale would throw

¹⁷MacroFinance/docs/latex-papers/docs/fable-rewrite/, especially findings.md, opinion.md, and rewrite-program-v1.md.

away real content, so the ZLB repair converted them: prohibition becomes consequence, negative definition becomes mechanism, and the accounting moves to an internal sidecar. The editorial test that separated the two cases, reason or protection, is implementable as a review prompt.

4. *Templates have ceilings.* The uniform section shape that the ZLB owner flagged was mandated by the project’s own style contract, and every checklist the BGS project introduced was promptly satisfied without the quality it named. Floors get gamed and become ceilings. Pattern lists and contracts therefore enter *humanvoice* only as subordinate diagnostics, and every diagnostic reports rather than auto-fixes, so that no rule can silently become the new template.
5. *Local repair does not compose.* The BGS volume absorbed 815 paragraph repairs and still had no accepted reader path; the ZLB rescue needed a register rewrite, not better sentences, before sentence work meant anything. The unit of failure is the document’s dependency graph. Sentence diagnostics are still worth building, but nothing at sentence granularity can certify a document, which is why the acceptance layer in Section 6 is reader-based.
6. *Self-review cannot certify a human voice, and context-sharing reviewers are contaminated.* Agent reviewers who shared the repository accepted what a human rejected; the context-starved persona review predicted the human. The design consequence is a review architecture: reader-proxy reviews run with no repository context, forced choices, and an instruction-injection boundary, and the human read remains the only promotion gate.
7. *Both failure directions exist, so length is not a signal.* Documents in this record were rejected for being compressed summaries (the BGS 21-page synthesis, the 23-page V11) and for being sprawling logbooks (snapshot 06, the 115-page bridge edition); SMEwallet’s accepted chapter was longer than its rejected drafts. “Make it shorter” and “make it longer” are both category errors; the target is selection and recomposition with the evidence layered elsewhere.
8. *Two documents, not one.* Every attempt to make one artifact serve as both audit record and human manuscript produced the hybrid readers rejected. The internal record keeps ledgers, anchors, and identifiers; the manuscript keeps none of them; an explicit identifier map ties them together. The ZLB rescue kept the internal markdown as the record and the LaTeX as the manuscript, with the code-name-to-math-symbol map written into the contract.
9. *Measured diagnostics changed outcomes; unmeasured review did not.* The ZLB repairs converged because someone counted 133 dashes and 20 identical openers, repaired, and recounted; ten unmeasured BGS review rounds added nothing. Numbers also disciplined the arguments about whether a repair had regressed. Every *humanvoice* diagnostic therefore emits counts with locations, and acceptance tests for the tools are replays of these recorded numbers.
10. *Guard the baseline or lose the right to edit aggressively.* Byte-identical equation bodies, label and citation censuses, and explain-every-removal reports are what let the ZLB editors rewrite freely without anyone fearing silent damage; SMEwallet’s protected-source maps

did the same job. The principle underneath is worth stating: readability improvement is never established by shortening, so every removal must be accounted for.

One warning belongs beside the list rather than in it. The BGS project ended up better at describing its failures than preventing them, and rewarding honest self-assessment gave its author an exit ramp. `humanvoice` should expect the same pull and should judge itself by one number only: how many rounds of human feedback a document needs before acceptance. This survey, whose first version failed that test immediately, is not exempt.

3 The tools we already have

Our answer to the failure record so far is a doctrine stack: one policy, three skills, one vendored pattern list, two grounding tools, and a distribution mechanism. This section reviews each piece the way a designer needs it reviewed: what it is, how it actually works, what evidence exists that it helped, what its merits are, and where it falls short. The summary judgment, argued piece by piece below, is that the doctrine layer is mature and consistent while three things are missing entirely: executable measurement, behavioral evidence, and a single home.

3.1 The prose policy

The constitution of the stack is the “Reader-Facing Scientific Prose” section of the global agent policy, roughly 150 lines inside the 618-line policy file installed into every agent’s context.¹⁸ Its load-bearing device is a precedence order. When rules conflict, an agent resolves them in this sequence: mathematical and scientific correctness first, then source and evidentiary fidelity, then domain-appropriate meaning, then reader comprehension and natural expression, and only then template regularity. The order is what keeps a pattern list from doing damage. A rule like “remove em dashes” sits at the lowest level, so it can never justify deleting a numeric range from a theorem statement; a readability preference can never justify dropping an assumption.

Three of its rules encode the failure record directly. The register rule forbids software and workflow vocabulary as metaphor in exposition: “artifact,” “pipeline,” “gate,” and “handoff” are reserved for actual software, which is precisely the administrative-register defect of Table 1 written as a prohibition. The rigor rule demands positive statements over defensive ones, with a worked example: prefer “We assume log utility in this example because it keeps the first-order condition simple” to “We are not claiming that households literally have logarithmic utility.” That is the defensive-qualification repair from Section 2.2.2, generalized. And the acceptance rule states that “author self-review, model review, policy installation, checklist completion, a style score, or successful compilation cannot certify a human voice,” which Section 2.6 showed the hard way. Beside the prose section sits the non-evasion vocabulary: a claim is “correct,” “wrong relative to the stated target,” “unsupported,” “not checked,” or “heuristic only,” and softening words may not blur which one applies.

Merits: the policy is complete in the sense that every failure class in Section 2 has a corresponding clause, and the precedence order resolves the conflicts that sank naive rule-following elsewhere. Problems: it is prose all the way down. Nothing in it executes, so compliance is whatever the drafting agent happens to retain from 618 lines of context, and the

¹⁸`claudecodex/policies/global-scientific-coding-agent-policy.md`. The companion section “Plain Scientific Language And Non-Evasion” supplies the verdict vocabulary discussed below.

one thing the policy cannot do is notice that it is being ignored. The ZLB rescue’s third pass began with the admission that the policy “had not been consulted.” A policy that must be remembered is a policy that will sometimes not be.

3.2 The narrative skill

`write-linear-scholarly-narrative` is the document-architecture skill, distilled from the SMEwallet Draft 11 method.¹⁹ It instructs an agent to do, in order, the things Draft 11 did. Freeze a human contract first: the target reader, their prerequisites, and what they must be able to explain afterward without hunting elsewhere. Design the plot before the contents, choosing the smallest narrative spine that makes the argument accumulate, and deriving chapter order from explanatory dependencies rather than from the source material’s order. Storyboard one bounded unit at a time, recording what the unit reactivates, the one question it opens, the smallest new idea, and the tension carried forward. Draft in a teaching order: eleven steps from “return to the running case” through “introduce notation only after the need is felt” to “make the next question feel necessary.” Then validate by reader reconstruction: a human reads the unit and must be able to rebuild why it began, every turn, what the equation was for, and why the next unit follows.

Its most valuable property is self-awareness about Section 2.6’s fourth lesson. Nearly every rule carries an anti-template clause; the drafting sequence is offered “as a menu,” with the instruction not to fill slots mechanically and not to add a caveat or transition merely because the template has a place for one. The working-memory budget (about four live conceptual items) is labeled a diagnostic threshold, not a law. Whoever wrote it had watched a floor become a ceiling.

Evidence of effect is real but bounded: Draft 11’s units passed their PI gates one by one, which no earlier method achieved. Problems: the skill has no examples inside it, so an agent meets its abstractions (“narrative spine,” “reactivate”) without a single worked instance; it has no tooling, so storyboards and contracts live or die by agent diligence; and its acceptance step needs a human per unit, which priced it out of the BGS lane’s volume. Nothing tests whether an agent following it writes measurably better than an agent that has never seen it, a gap Section 7’s third work package exists to close.

3.3 The sentence-level skill

`write-natural-analytical-prose` is the sentence-and-paragraph companion.²⁰ Its core is a two-sided safety mechanism. Before any rewrite, the agent freezes a meaning ledger: the facts, attributed claims, deductions, live uncertainties, author judgments, and exact numbers that must survive. After the rewrite, it diffs the result against the ledger, and a smoother sentence that changed a claim counts as a failure, not an improvement. Between those two steps sit the craft rules: give each sentence one epistemic job, with a calibrated verb table in which “shows” is not a synonym for “suggests” and “reproduces” is not a synonym for “validates”; put concrete actors in subject position (“the regime test rejects the authors’ path,” not “the evidence indicates”); state the finding before its qualifications; and spend each qualification

¹⁹`claudecodex/skills/write-linear-scholarly-narrative/SKILL.md`, with a 259-line companion, `references/teaching-contract.md`, defining the comprehension standard.

²⁰`claudecodex/skills/write-natural-analytical-prose/SKILL.md`, with before/after contrasts in `references/prose-patterns.md`.

once, at the point of likely misunderstanding, rather than re-attaching it to every positive sentence.

This is the skill that operationalizes the epistemic-blur and abstract-actor rows of Table 1, and its meaning-ledger loop is the sentence-scale version of the protected-baseline invariant. Its problems mirror the narrative skill’s: no automation, so the ledger diff is manual; verification by “read it aloud”; and hard prohibitions that depend entirely on the drafting agent noticing its own violations. The skill is also, like everything in this stack, tested only by phrase-lock (see Section 3.8): the test suite asserts that certain sentences still appear in the skill file, not that the skill changes any output.

3.4 The humanizer pattern list, as installed

On 20 August the workspace vendored `blader/humanizer` v2.11.2 into the policy tree, as `humanizer-ai-writing-patterns.md`.²¹ The list itself is 35 numbered AI-writing patterns derived from Wikipedia’s “Signs of AI writing” catalogue, each with words to watch and a before-and-after pair; Section 5.2 walks through its content. What matters here is the installation decision. The file carries a locally written precedence header that demotes the whole list to “a DIAGNOSTIC pattern list, subordinate to global-scientific-coding-agent-policy.md,” spells out the order of authority, and singles out the one rule that would otherwise do damage: the flat dash ban of its pattern 14 is to be applied “proportionately rather than as a ban,” and no pattern may be satisfied “by deleting mathematics, assumptions, or qualifications.”

That header exists because of a live incident. During the ZLB rescue the list was consulted without a precedence frame, and the case study records the realization that without one, “a pattern list becomes a new template and you trade one uniformity for another.” The header is currently the only policy-precedence mechanism in the workspace, and it is a comment in a markdown file. It governs an agent exactly as far as the agent reads and honors it. Turning this from a comment into configuration, per-format rule downgrades in the diagnostic layer itself, is one of the cleaner design requirements to come out of this review (Section 6, R2).

3.5 The repair skill and the only working scripts

MathDevMCP ships its own scholarly-writing skill, `repair-scholarly-readability`, and it is the only piece of the stack with executable parts.²² Where the narrative skill teaches drafting, this one teaches repair of an existing LaTeX document, and its structure encodes two ideas the rest of the stack lacks. The first is four separate ledgers: reader comprehension, mathematical integrity, source fidelity, and typography, with the explicit rule that “evidence in one ledger cannot certify another.” A clean compile is typography evidence only; a passed equation audit says nothing about motivation. The second is bounded repair: one mechanism chapter or 6–12 rendered pages at a time, against a preserved baseline, ending in a cold-reader test with prespecified questions and a four-ledger decision table.

Both scripts matter beyond their modest size. `scripts/scholarly_surface_audit.py` emits JSON and markdown surface diagnostics for a `.tex/.pdf` pair, and `scripts/render_review_pages.sh` renders page images for visual inspection, institutionalizing the ZLB rescue’s

²¹`claudecodex/policies/humanizer-ai-writing-patterns.md`. Section 5.2 reviews the upstream project; this subsection reviews our installation of it.

²²`MathDevMCP/skills/repair-scholarly-readability/SKILL.md`, with three reference rubrics and two scripts under the same directory.

discovery that text extraction lies about layout. These are the embryo of the `hv gate` command in Section 7.

Evidence of effect: the skill’s bounded method took the opening of the BGS industry-DSGE dossier through a repair that passed five prespecified cold-reader questions, the strongest automated-plus-human result in the BGS lane. The same result file records the limit: later chapters “remain dry,” and the project’s audit concluded the skill “generalizes locally, not across a corpus-sized argument,” which is lesson five of Section 2.6 observed in the field. The other problem is organizational: this skill overlaps the claudecodex narrative skill in scope while living in a different repository with different reference rubrics. Scholarly readability currently has two homes and no owner.

3.6 ResearchAssistant

ResearchAssistant is the source-grounding half of citation discipline: a local-first tool that ingests papers (preferring arXiv LaTeX source over PDF, because the LaTeX parses reliably and the PDF does not), extracts sections, equations, labels, and bibliographies, tracks provenance and review status, and exposes a read-only MCP surface plus a CLI.²³ For this survey its relevant property is precedent: its `ra survey` subcommands are the one place where a piece of workspace writing policy became software. The literature-audit policy demands six ledgers (source support, citation metadata, backward and forward snowballing, claim support, omitted-paper risk); `ra survey central-papers` writes exactly those ledgers and returns per-paper dispositions.

Its behavior under failure is the part worth copying. In the ZLB campaign the tool’s public-discovery bootstrap was unavailable, and instead of degrading silently it closed the mission as `terminal_blocked_bootstrap_unavailable`, leaving the ledgers to be maintained by hand. A tool that reports its own incapacity honestly is rarer than it should be. Its limits for `humanvoice` are equally clear: it grounds sources, not prose; its equation extraction from PDFs is explicitly marked unreliable (arXiv LaTeX is the reliable path); and its discovery pipeline depends on external metadata services that were down when most needed.

3.7 The mathematics side: MathDevMCP and DynareMCP

Two more tools border the writing problem without being writing tools. MathDevMCP’s mathematical layer provides SymPy-backed equality checking, derivation audits, and an experimental whole-document rigor audit over LaTeX. Its one recorded contribution to a manuscript is instructive about scale: the ZLB audit produced per-equation issue ledgers and exactly one applied exposition patch (an invertibility condition stated before a displayed inverse), tagged “candidate_exposition_patch_not_certificate.”²⁴ That is the correct division of labor, mathematics tools nominating exposition fixes without certifying prose, and `humanvoice` should preserve it rather than absorb it. DynareMCP, the macro-model execution server, contributes structure inventories that were used to falsify inflated rewrite claims (counting inherited words and relocated blocks), a niche but genuinely useful falsification tool for “we rewrote it” assertions.

²³`research-assistant/`; the checked-in MCP configuration targets Claude Code and VS Code. Python 3.11 only.

²⁴`BayesFilter/docs/surveys/zlb_discontinuous_hmc/mathdevmcp_audit_20260819.md`.

3.8 Distribution, and what the tests actually test

claudecodex is the delivery mechanism. Its installer splices the global policy into marked blocks in each agent’s configuration files idempotently, merges approval rules, and copies the skills into the Codex skill tree.²⁵ Around it sit the cross-agent review plumbing (bounded review packets, probe ladders, a gate script that parses “VERDICT: AGREE or REVISE”) and a five-file proposal template system with seven evidence ledgers. The machinery works and is in daily use.

Its test suite deserves a hard look because of what it teaches about the stack’s blind spot. There are real behavioral tests for the review gate (a fake worker exercises the probe-and-fallback paths) and for the installer. For the writing skills there are only phrase locks: `test_natural_analytical_prose_skill.py` asserts that certain sentences still occur in the skill file and that banned phrasings do not. Phrase locks prevent silent regression of the doctrine’s text. They say nothing about whether the doctrine changes what an agent writes. As of today, no experiment anywhere in the workspace has compared documents produced with and without any of these skills. The doctrine is plausible, consistent, and untested.

Two further gaps are administrative but consequential. Installation is Codex-first; Claude Code agents receive the policy but not a native skill installation. And the writing doctrine now lives in three places, the claudecodex policy tree, the claudecodex skills, and the MathDevMCP skill, with cross-references in prose instead of a shared source of truth.

3.9 The gap, stated precisely

Table 2 compresses this section. Reading its last column top to bottom gives the build list: the six diagnostics the ZLB case study asked for exist nowhere as code (register linter, rhythm profiler, defensive-register finder, LaTeX-aware baseline differ, first-class two-document support, and a precedence mechanism that is configuration rather than a comment); the doctrine has no behavioral evidence; and consolidation is overdue. That is what Sections 6 and 7 are for.

4 What the research literature says

This section explains the studies a designer of `humanvoice` needs to understand, at the level of what the researchers actually did and found. Each subsection ends with the consequences for our design, so that Section 6 can cite conclusions rather than re-argue them. A note on inspection depth: the load-bearing papers below were read at method level (design, data, and headline numbers verified from the full text or extended abstract); the supporting papers were read at abstract level and are cited for orientation only. One result, StoryScope, reached us secondhand through an open-source project and is marked accordingly wherever it appears.

4.1 The fingerprint: what makes text read as machine-written

Start with the study that made the phenomenon measurable. Kobak et al. [2025] asked how much of the biomedical literature is now LLM-assisted, without using any detector. They parsed more than fifteen million English PubMed abstracts from 2010–2024 and tracked, for

²⁵`claudecodex/install_global_agent_policy.py`; templates and review-gate scripts under `claudecodex/docs/templates` and `claudecodex/scripts`.

Piece	What it reliably provides	What it cannot do
Prose policy	complete doctrine; precedence order	execute, or notice being ignored
Narrative skill	Draft-11 method; anti-template clauses	show examples; run without a human per unit
Sentence skill	meaning-ledger safety loop; verb calibration	automate the ledger diff
Humanizer (installed)	35-pattern diagnostic with FP guards	bind its own precedence outside prose
Repair skill	four-ledger repair; the only scripts	compose beyond 6–12-page units
ResearchAssistant	source grounding; ledgers as software	anything about prose
MathDevMCP/DynareMCP	equation audits; structure inventories	certify exposition
claudecodex	idempotent install; review plumbing	test writing behavior; reach Claude natively

Table 2: The existing stack, by what each piece does and does not provide.

every word, the fraction of abstracts containing it each year,

$$p_t(w) = \frac{a_t(w) + 1}{b_t + 1}, \quad (1)$$

where $a_t(w)$ counts abstracts in year t containing word w and b_t is the year’s abstract total. Borrowing the excess-mortality design from epidemiology, they project a counterfactual 2024 frequency from the pre-LLM trend,

$$q_{2024}(w) = p_{2022}(w) + 2 \max(p_{2022}(w) - p_{2021}(w), 0), \quad (2)$$

that is, the 2021→2022 change extrapolated two years forward and clamped so the counterfactual never falls (2023 is skipped as already contaminated). A word is in excess if its gap $\delta = p - q$ or its ratio $r = p/q$ clears a threshold. The result that matters for us is *which* words were in excess. The Covid years produced excess content words (nouns about the pandemic); 2024 produced excess *style* words: “delves” at a ratio near 28, “underscores,” “showcasing,” “potential” with a gap of 0.045. From the share of abstracts containing at least one excess style word, they bound LLM assistance below by 13.5% of 2024 abstracts, up to 40% in some subcorpora. The bound is a floor, since authors who scrub the telltale words become invisible, and that caveat cuts both ways for us: scrubbing words defeats this measurement without improving the prose.

Why do models overuse exactly these words? Juzek and Ward [2025] isolated the vocabulary by a three-way intersection: words whose PubMed frequency jumped significantly between 2020 and 2024, that ChatGPT also overuses when asked to write abstracts for 2020 papers (so the comparison is like-for-like), yielding 21 focal words. They then tested explanations. The words are not overrepresented in plausible training corpora, which weakens the training-data story. The sharpest test compares Llama-2-7B Base with its Chat variant, identical architecture, the Chat model adding only supervised fine-tuning and RLHF: the Chat model is markedly less surprised by focal-word-laden text. The overuse enters at the human-feedback stage. One more of their findings matters for design: raters recruited to resemble the RLHF workforce showed

no overall preference *for* the focal words, so nothing in the reward signal pushes back against amplification once it starts.

For practitioners, the catalogue of the fingerprint is Wikipedia’s “Signs of AI writing,” maintained by the editors who clean AI text out of Wikipedia [WikiProject AI Cleanup, 2026]. It goes well beyond vocabulary: inflated significance (“stands as a testament”), superficial -ing analysis tails, rule-of-three padding, negative parallelism (“not just X, but Y”), formulaic challenges-and-future sections, essay-like conclusions, title-case headings, and chatbot residue. Two properties make it the right seed list for diagnostics. It is maintained against live cleanup experience, so it updates as the models change; and its preamble insists that no single sign is diagnostic, since human writers produce all of them at some rate. That warning is the germ of the false-positive discipline that the best open-source tools in Section 5 implement and the worst ignore.

What about “slop” as a quality construct rather than an authorship one? Shaib et al. [2025] asked nineteen experts across NLP, writing, journalism, linguistics, and philosophy to define it, coded the definitions, and landed on three themes: information utility (density and relevance), information quality (factuality, bias), and style quality (repetition and templatedness, coherence, tone). Professional copy-editors then annotated news and retrieval-augmented QA passages, first with a binary slop judgment, then with word-level spans tagged by code. Two findings carry over to us. Regressions showed *relevance and density*, not style, best predict the binary slop judgment, which is a formal version of our record’s lesson that substance failures hide under style failures. And even trained annotators agreed only moderately (theme-level Krippendorff’s α of 0.34–0.45), which calibrates how much precision we should expect from any judge, human or model, on this construct.

Finally, homogenization. Padmakumar and He [2024] hired 38 writers to produce short argumentative essays in three within-subject arms: alone, with base GPT-3 suggestions, and with feedback-tuned InstructGPT suggestions, with keystroke logging attributing every character to writer or model. Essays co-written with the instruction-tuned model were measurably more similar to one another (higher pairwise Rouge-L and BERTScore within topics, fewer distinct n-grams and key points), the base model showed no such effect, and the decomposition traced the homogenization to the model-contributed text specifically. The instruction tuning, not the assistance, is the homogenizer. Doshi and Hauser [2024] found the same shape in a creative-writing field experiment, individual quality up, collective diversity down, and Guo et al. [2023] showed the spiral direction: models trained recursively on synthetic text lose lexical, syntactic, and semantic diversity generation over generation. One discourse-level result is worth logging despite its secondhand status: the StoryScope study is reported to detect machine narrative at 93% F1 from discourse features alone, with professional surface rewriting moving detection by less than two points [Russell et al., 2026]. We have not inspected the paper; if it holds, it independently confirms that the durable fingerprint is structural, which is also what our failure record found when 815 sentence repairs left a volume unaccepted.

Consequences for the design. The fingerprint is distributional and structural, not a word list. So *humanvoice* measures excess against a genre reference corpus, Kobak-style, with equations (1)–(2) applied to pre-2022 economics text rather than PubMed; it treats word lists as seeds for measurement, not rewrite targets; it measures homogenization *across* our documents, not only within one; and it expects style repair alone to leave the deeper fingerprint intact, which is why the acceptance layer is reader-based.

4.2 Why the models write this way

Four studies, read together, explain the register our agents default to, and the explanation changes what interventions can work.

Kirk et al. [2024] trained the same base models through the alignment stages, supervised fine-tuning, best-of- n sampling, and PPO-based RLHF, on summarization and instruction-following, then measured performance (GPT-4-judged win rates, in and out of distribution) and output diversity (distinct n -grams, embedding dispersion, and an entailment-based measure, per input and across inputs). RLHF wins on performance, in and out of distribution, and pays for it with a substantial drop in per-input diversity. Notably, turning the KL penalty up, the knob that is supposed to keep the tuned model close to the base model, lowered performance without restoring diversity. The diversity loss is not a mis-set hyperparameter; it is what the objective buys.

Why does preference optimization sharpen so hard? Zhang et al. [2025] supply the cleanest mechanism: typicality bias in the preference data itself. Human raters, for reasons cognitive psychology has documented for decades (mere exposure, processing fluency), systematically favor familiar, schema-congruent text. Model the reward as true utility plus $\alpha \log \pi_{\text{ref}}(y|x)$, a bonus for being typical under the reference model. On 6,874 preference pairs whose two responses are *equally correct*, the fitted α is about 0.57–0.65 and highly significant: raters reward typicality even with correctness held fixed. Optimizing that reward provably sharpens the policy toward $\pi_{\text{ref}}^{1+\alpha/\beta}$, collapsing onto the mode whenever many answers tie on utility, which is precisely the situation in prose style. Their intervention is a prompt change: ask for *five responses with their probabilities* rather than one response. Requesting a verbalized distribution recovers 1.6–2.1 $\times$ the diversity of direct prompting on creative tasks and about two thirds of the base model’s diversity across post-training checkpoints, at no retraining cost.

Verbosity has its own paper trail. Singhal et al. [2023] decomposed where RLHF’s reward gains come from on three preference datasets, and the answer is mostly length. Within fixed length bins, reward improves little (on one dataset, within-bin gains are a few percent of the total); shifting mass toward longer outputs accounts for 70–90% of the improvement. Their sharpest demonstration replaces the learned reward with a pure length score: PPO against “longer is better” reproduces most of the measured win-rate gain of PPO against the real reward model. Padding is not a stylistic accident; it is what the training objective paid for. Sharma et al. [2024] complete the picture on the agreeable side: preference-trained assistants systematically flatter and hedge because raters prefer agreement, the same mechanism that seeds throat-clearing and deference into technical prose.

Consequences for the design. These are training-time artifacts, so prompt-level pleading (“write naturally”) fights the model’s objective and loses; what works is changing the sampling question (verbalized sampling), editing after generation, and measuring. Compression deserves first-class status among edit operations, since verbosity is a paid-for bias. And any LLM used *inside* humanvoice as an editor or judge carries these same biases, which is why the judging protocol below is designed around them rather than around trust.

4.3 Detection: how it works and why it cannot be a gate

humanvoice needs to understand detectors for two reasons: their failure modes rule out an entire class of tempting designs, and their internal statistics, used gently, are legitimate population-level

instruments.

The detection literature is one idea refined three times. Machine text sits where the model expects text to sit; human text does not. GLTR [Gehrmann et al., 2019] visualized this directly, coloring each token by its rank in the predicted distribution, and lifted human detection accuracy from 54% to 72% just by showing the ranks. DetectGPT [Mitchell et al., 2023] made it a statistic: perturb the passage (mask spans, let T5 refill them, about a hundred variants), score original and variants under the suspected source model, and flag when the original sits at a local likelihood peak, since sampled text lies at peaks and human text does not. It works (0.95 AUROC on their news benchmark) but needs roughly the right source model and a hundred model calls per document. Fast-DetectGPT [Bao et al., 2024] removed the rewriting: one forward pass gives the conditional next-token distribution at every position, and the statistic compares the log-probability of each actual token with the mean and spread of alternatives sampled from that same conditional,

$$\text{curvature} = \frac{\log p(x) - \tilde{\mu}}{\tilde{\sigma}}, \quad (3)$$

with machine tokens scoring near the top of their conditionals (curvature around 3) and human word choices looking like ordinary draws (around 0). A surrogate scorer makes it black-box, it is 340× faster than DetectGPT, and it flags 80% of ChatGPT passages at a 1% false-positive rate on benchmarks. Binoculars [Hans et al., 2024] normalizes differently: the ratio of a passage’s perplexity under one model to its cross-perplexity between two sibling models, which cancels the “weird prompt” confound (their copybara problem) and detects across generators with one global threshold, over 90% detection at 0.01% FPR on their benchmarks.

Now the failures, which are the design-relevant part. Liang et al. [2023] ran seven widely used detectors on two human corpora: essays by US eighth-graders and TOEFL practice essays by non-native speakers. The US essays passed nearly clean; the TOEFL essays were flagged as AI at an average false-positive rate of 61%, with 98% flagged by at least one detector. The mechanism is exactly the detectors’ core statistic: limited lexical range produces low perplexity, and low perplexity reads as machine. Two prompt experiments nailed it: asking ChatGPT to enrich the TOEFL essays’ word choice dropped their false-positive rate to 12%, and asking it to simplify the US essays raised theirs to 57%. Polished, formal, conventional prose, which is to say *scholarly* prose, sits on the wrong side of this statistic by construction. Weber-Wulff et al. [2023] tested fourteen academic and commercial tools and found none reliable, with accuracy below 80% and easy degradation; Krishna et al. [2023] showed that one round of automatic paraphrase defeats the perturbation-family detectors wholesale. Detection is unreliable exactly where we would use it and trivially evaded by anyone trying.

There is one statistically respectable use, and it is population-level. Liang et al. [2024a] estimate the *fraction* α of a corpus that is substantially LLM-modified without classifying any document. Represent each document by which adjectives it contains; estimate each adjective’s occurrence probability under a human reference corpus (P , pre-ChatGPT reviews) and an AI reference corpus (Q , generated reviews); then choose α to maximize the mixture likelihood

$$\hat{\alpha} = \arg \max_{\alpha} \sum_i \log((1 - \alpha) P(x_i) + \alpha Q(x_i)). \quad (4)$$

On constructed mixtures with known α the estimation error is under two percent, an order of magnitude better and about 10^7 times cheaper than aggregating per-document detectors.

Applied in the wild: 10.6% of ICLR 2024 review text substantially modified (up from 1.6%), 16.9% at EMNLP, no change at Nature-family journals, and higher α in late, low-confidence reviews. The estimator is meaningless for accusing an individual document, by construction, which is exactly the property we want.

Consequences for the design. No per-document authorship verdict, ever: the false-positive profile lands on precisely the writers and the register we care about, and evasion is cheap anyway. What survives is (i) curvature or Binoculars-style scores as *corpus-level* smoothness trends, advisory only, computed on detexed prose; and (ii) the mixture estimator of equation (4) as the statistically honest way to answer “how much of our output still reads machine-typical this quarter,” with P estimated from pre-2022 economics text and Q from our own agents’ raw drafts. Both enter `hv drift` in Section 7, gated against per-document use.

4.4 The edit paradigm: the strongest positive evidence

Most directly useful for humanvoice is Chakrabarty et al. [2024a], because it tests the exact pipeline we propose: generate, diagnose, edit. They generated about a thousand paragraphs with three frontier models from instructions reverse-engineered out of published prose, had professional editors mark them up, and consolidated the editors’ labels into seven idiosyncrasy categories: cliché, unnecessary or redundant exposition, purple prose, poor sentence structure, lack of specificity and detail, awkward word choice, and tense inconsistency. Eighteen professional writers then produced 8,035 categorized span-level edits, the LAMP corpus. Three of their findings set our expectations. Writing quality did not differ across the three models, so the idiosyncrasies are an ecosystem property, not a vendor one. An LLM-based span detector reached 0.46 precision against expert spans, but the experts themselves only agreed at 0.57, so “flag the bad spans” is inherently noisy for everyone and should feed a human or model editor rather than an automatic rewrite. And the payoff experiment: in blind rankings by the writers, edited AI text (whether the spans came from experts or the detector) beat raw AI text decisively, raw output ranked last 60% of the time, while human-edited text still ranked first. Editing recovers most of the gap; it does not close it.

Two smaller results refine the objective. Cheng et al. [2025] built a corpus-based metric of human-like tone and a steering method, and found that users do not uniformly prefer maximal human-likeness: the optimum depends on genre and task. “As human as possible” is the wrong target; “appropriate to the register” is the right one, which for us is fixed by the field’s own literature. And the style-transfer literature [Jin et al., 2022] is the cautionary map: pre-LLM transfer methods chronically traded content preservation against style strength, which is the trade our meaning-ledger and baseline invariants exist to refuse.

Consequences for the design. Build the editor as span-diagnose-then-edit with categories, not as one-shot rewriting; expect span precision near 0.5 and design the workflow so a human or a second model adjudicates flags; measure edit effect by blind pairwise comparison; and never let an edit change a claim, which our existing meaning-ledger skill already enforces in doctrine and `hv diff` will enforce in code.

4.5 Judging writing with models, without fooling ourselves

humanvoice needs cheap judgments of “which version reads better,” so the LLM-as-judge literature is load-bearing, mostly for its list of traps.

Zheng et al. [2023] established the baseline: on their multi-turn benchmark, GPT-4’s agreement with pooled expert votes was 85%, against 81% human-human agreement, so a strong model is a usable proxy for preference between chat answers. The same paper measured the biases operationally. Position bias: judge two near-identical answers twice with order swapped; every judge showed order-dependent flips. Verbosity bias: pad one answer with rephrased duplicates of its own list items, adding no information; two of three judges preferred the padded answer more than 90% of the time. Self-enhancement: models rate their own outputs higher, though their design could not settle causality. Panickssery et al. [2024] settled it with a clean causal design on summarization: models recognize their own outputs above chance out of the box, and fine-tuning a model to be *better* at self-recognition increases its self-preference linearly, while control fine-tunes on unrelated tasks do not. Self-preference rides on self-recognition, so the generator must never judge its own rewrite. Dubois et al. [2024] showed the verbosity bias is largely removable by regression: length-controlled win rates change model rankings and nearly triple agreement with human evaluations on their benchmark. Finally, Chakrabarty et al. [2024b] calibrate optimism for prose specifically: expert writers applying a creativity rubric failed LLM-written stories on roughly ten times more rubric items than professional stories, while LLM judges rated the same stories generously. For writing quality, model judges drift lenient exactly where experts are harsh.

Consequences for the design. The judging protocol writes itself, negatively: pairwise, never absolute scores; order randomized or both orders averaged; length-controlled; a different model family from the generator and from the editor; one rubric dimension per call rather than holistic “is this good”; and periodic validation of the whole proxy against recorded human verdicts, which our evaluation corpus contains. Any result that survives those controls is worth acting on; nothing else is.

4.6 Readability science: measure against the genre, not a formula

Classical readability formulas are the wrong instrument for this project, and it is worth knowing why rather than just asserting it. Flesch-family scores are functions of syllable counts and sentence lengths. LLM prose often scores *well* on them while failing readers, because its failures live in cohesion, rhythm, and register, which the formulas do not see. The modern evidence: the CLEAR corpus work collected thousands of passages with human ease-of-reading judgments and showed learned models beat the classical formulas substantially [Crossley et al., 2022]; Coh-Metrix demonstrated two decades ago that texts separate on cohesion and discourse dimensions the formulas ignore [Graesser et al., 2004]; and Biber’s register work established that genres occupy different regions of a multidimensional feature space, so “natural” is genre-relative [Biber, 1988]. Two findings about scientific prose complete the picture. Scientific abstracts have been getting harder to read for a century, with jargon and sentence complexity rising steadily [Plavén-Sigra et al., 2017], so the median published abstract is a degrading target, not a gold standard. And jargon density carries a professional price: titles and abstracts heavier in specialized terminology collect fewer citations [Martínez and Mammola, 2021], a result an economist will recognize as an incentive argument for plain language, not just an aesthetic one.

Consequences for the design. Every humanvoice metric is referenced to a target-genre human corpus (for us: pre-2022 economics working papers and journal articles), never to an absolute scale; the diagnostic suite spans lexis, sentence rhythm, and discourse cohesion rather than syllable arithmetic; and “write like the median abstract” is explicitly not the objective, which the policy’s plain-language stance already reflects.

4.7 Economics and publishing norms

Norms here are settled enough to build on. Korinek [2023], the profession’s de facto guide, treats LLM writing assistance as ordinary productivity technology for economists, to be used and disclosed. Journal policy has converged on disclosure rather than prohibition (Nature portfolio: models cannot be authors, use must be documented; Science moved from ban to disclosure). The prevalence work says assistance is already everywhere: the mixture estimator of Section 4.3 puts substantial LLM modification at 7–17% of recent ML conference review text [Liang et al., 2024a,b]; AI-assisted reviews measurably shift scores and acceptance at ICLR [Latona et al., 2024]; adoption in research writing skews toward non-native English speakers and junior researchers, whose styles are converging [Lin and Zhu, 2025]; and undisclosed chatbot artifacts (“as an AI language model...”) appear verbatim in published papers [Glynn, 2024].

Consequences for the design. humanvoice is a quality project under disclosure norms, full stop. Detector evasion is not a goal, would not be an achievement (Section 4.3 showed evasion is cheap), and is ruled out as a success metric because detectors cannot define quality. The prevalence work also hands us our reference corpora: pre-2022 economics text is a clean human baseline; anything after 2023 is contaminated and unusable for calibration.

5 Open-source tools, reviewed

We evaluated fourteen open-source projects: the four named in the project brief and ten found by searching for AI-tell skills, prose linters, detectors, and LaTeX-aware writing tools. Every project was cloned and read at the level of its actual pattern lists, prompts, rules, and code, not its README; the runnable ones were run on our own documents. The clones are pinned under `humanvoice/vendor/` with a commit manifest.²⁶ This section reviews each tool in enough detail to defend its verdict: what it contains, how it behaves, what is good, what is wrong, and what adopting it would concretely mean. Table 4 at the end collects the verdicts.

5.1 A calibration experiment first

One experiment frames every review that follows. sloptrim (Section 5.7) is the best deterministic AI-tell scorer we found: 0–100, fully local, documented patterns. We ran it on three documents whose human verdicts we know. Table 3 shows the result.

Read the table twice. First reading: the scorer works on its own layer. It separated the ZLB before (45, driven by boldface density and inline header lists, the markdown-era register) from the ZLB after (25), in the right direction, deterministically, in about a second. That is real value: regressions on this layer become visible without a human. Second reading: the layer

²⁶`humanvoice/vendor/MANIFEST.md` records the twelve cloned repositories with their commit hashes as of 21 August 2026; two further tools (TeXtidote, textlint) were surveyed without cloning.

Document	Score	Band	Human verdict
ZLB survey, internal markdown (pre-rescue)	45	mixed	rejected (register)
ZLB survey, final LaTeX (post-rescue)	25	light tells	accepted
This survey, first version	10	clean	rejected (does not teach)

Table 3: sloptrim v0.9.2 scores versus human verdicts. Lower scores mean fewer mechanical tells. The scorer correctly separates the ZLB rescue’s before and after, and rates the rejected first version of this survey cleaner than the accepted manuscript.

is not the one that decides acceptance. The first version of this survey scored 10, the cleanest of the three, and was the most decisively rejected, because its failure, density without explanation, lives above the mechanical layer. This is our record’s layered-masking lesson (Section 2.6) reproduced by instrument. Every tool below should be read against this calibration: the question is never “does it certify quality” (nothing here does) but “which layer does it measure, and how honestly.”

5.2 blader/humanizer: the canonical pattern list

One markdown skill, 457 lines; MIT; 36.8k stars; active. Verdict: adopted, already installed; we are current with upstream.

humanizer is Wikipedia’s “Signs of AI writing” operationalized as an editing skill: 35 numbered patterns in six groups (content, language and grammar, style, chatbot artifacts, filler and hedging, plus late additions), each with words to watch and a before-and-after pair. The content patterns cover inflated importance, name-dropping, -ing-phrase pseudo-analysis, sales language, vague sources, and formulaic outlook sections; the style patterns cover dashes, bold density, list formatting, title case; the chatbot patterns cover residue like “I hope this helps”; the late patterns, 34 and 35, name two register defects our record knows well, answering objections nobody raised and rejecting alternatives nobody proposed, which makes the list unusually aligned with our defensive-qualification class.

Two design features separate it from its many imitators. It has a serious false-positive section, “What not to flag”: em dashes alone prove nothing, formal vocabulary is not a tell, scope statements and genuine disclaimers stay, secondhand text (quotes, titles, discussed examples) is never rewritten. And it has a fact-preservation rule: the rewrite may not add or drop a claim, number, date, quote, or citation, with any unsupported addition counted as an error. Those two sections are why our installation (Section 3.4) could be made safe with one precedence header; a list without them cannot be made safe at all.

Its one dangerous rule is pattern 14: a flat ban on em and en dashes unless the writer’s sample uses them, with a mandated final search for both characters. Applied to scholarly LaTeX it would strip mandatory typography, en dashes in numeric ranges and name compounds (“Nash–Cournot”). Our precedence header already overrides it to proportionate application; the configuration layer of Section 7 will enforce the override mechanically. We diffed our vendored copy against upstream: upstream `main` is at the same content version (2.11.2; the version bump since the last tag was a README-only change), so no update action is needed.

5.3 hardikpandya/stop-slop: sharper, narrower, wrong register

One 68-line skill plus three reference files; MIT; 16k stars; last substantive commit March 2026. Verdict: mine for patterns; do not install.

stop-slop is an essayist’s de-slopping skill: eight core rules, a quick checklist, and a five-dimension 1–10 self-scoring rubric (directness, rhythm, trust, authenticity, density; below 35 of 50, revise). Its distinctive material is genuinely distinctive. *False agency*: “the data tells us,” “the decision emerges,” an inanimate subject performing a human verb where a person should be named. *Negative listing*: “It wasn’t X. It wasn’t Y. It was Z,” the multi-sentence teaser stack. *Narrator-from-a-distance* and the interrogative-opener crutch. None of these appear in humanizer, all of them appear in our agents’ drafts, and all three transfer to our catalogue, with one carve-out: “the data tell us” and “the model implies” are ordinary, correct econometric usage, so the false-agency rule needs a domain exemption list before it goes anywhere near a manuscript.

Installing the skill itself is ruled out by its register assumptions. “Kill all adverbs. No -ly words” would delete “approximately,” “asymptotically,” and “conditionally” from an econometrics paper. “No passive constructions” conflicts with methods prose where the actor is irrelevant by design. “‘You’ beats ‘People’” and “put the reader in the room” import a blog voice that scholarly writing correctly refuses. “Two items beat three” and the ban on “every/always/never” collide with quantifier language. These are not flaws for its intended genre; they are a genre boundary, and it is on the wrong side of ours.

5.4 leonxlnx/taste-skill: not a writing tool

Thirteen skill files, 6,670 lines; MIT; 78.6k stars; active. Verdict: skip.

Despite surfacing in every search for writing skills, taste-skill is a frontend design pack: landing pages, brand kits, image-generation prompting, redesign checklists. The writing content it does contain is marketing-copy hygiene for web pages (register locks for button text, bans on fake-precise numbers and “Quietly trusted by”), not prose craft. We keep it in the record for one sociological observation. Its em-dash rule is an absolute zero, and the maintainer’s comment explains why: the agent “historically ignored em-dash limits when phrased as ‘use sparingly.’” Proportionate instructions decay into non-compliance; flat bans survive because they are checkable. That is an argument for putting proportionality into a machine-checked configuration (where “at most n per section, except in math” is enforceable) rather than into prose instructions, and it is the last time we will be surprised that every popular skill bans dashes outright.

5.5 itallstartedwithaidea/writing-agent: rejected on both counts

Node.js multi-agent framework with a 40-check QA runner and a 200-phrase blacklist; MIT; 20 stars. Verdict: skip.

writing-agent is the one evaluated project whose goal conflicts with ours. Its framing is detector evasion: checks are named “Watermark Stripping,” “Model Attribution Evasion,” “False Positive Rate Exploitation,” and “Non-Native Bias Exploit,” the last being the deliberate exploitation of the bias documented by [Liang et al. \[2023\]](#). Section 4.7 settled our position: quality under disclosure norms, never evasion, so this orientation alone disqualifies it as a dependency.

Its engineering would disqualify it anyway, and it is worth one example because “40-point QA” sounds substantial until read. Check 19, “Watermark Stripping,” begins with the comment: “SynthID watermarks are token-level statistical patterns. We can’t detect them directly, but we check that content was substantially rewritten,” and returns a pass with an aspirational message. Roughly a third of the forty checks are of this kind, unconditional passes wearing serious names. The “perplexity” check is an uncommon-word ratio against a 200-word stoplist. The blacklist flat-bans “moreover,” “furthermore,” and “in conclusion,” legitimate scholarly connectives whose clustering, not existence, is the tell. The few sound numeric ideas inside (sentence-length standard deviation targets, paragraph-length variation, type-token ratio) exist in better-engineered form in sloptrim and vale-ai-tells. Nothing here needs mining.

5.6 Vale plus tbhb/vale-ai-tells: the CI measurement host

Vale: markup-aware prose linter in Go; MIT; 6k stars; very active. vale-ai-tells: a Vale rule package, 111 prose rules, 15 commit-message rules, 18 experimental structural metrics; MIT; 73 stars; active, with a false-positive test corpus in CI. Verdict: adopt with a curated configuration.

Vale is the standard way to run prose rules in CI: rules are YAML files grouped into styles, findings come back with file, line, severity, and count, and per-format configuration can enable, disable, or downgrade any rule for any file glob. That last property is the point for us. Our policy-precedence principle, “diagnostics subordinate to domain correctness,” becomes an ordinary configuration file: a rule that is wrong for LaTeX is switched off for `*.tex` without touching the rule.

vale-ai-tells is the rule package that makes Vale worth adopting now, and its engineering quality is visible in the rule bodies. Three examples show the range. A vocabulary-class rule is a token list with a message; a structural rule is a formula; a defensive-register rule is a set of shape patterns. Respectively:

DefensiveHedges.yml: flags shapes like “This is more of X than Y, but...,” “It’s not the most X, but...,” “This may seem X, but...” with the message “State your point directly without pre-defending it.”

AverageSentenceLength.yml: a metric rule whose formula is words over sentences and whose condition is “greater than 25,” with a message noting AI text tends toward uniformly long sentences.

SentenceStartEntropy.yml: `extends: script`, running a small Tengo program that computes the entropy of sentence-opener distribution per section and warns on monotony.

Where the package earns adoption is its experimental style, the measurable-diagnostics collection our case study asked for: sentence-length variance, sentence-start entropy and repetition, paragraph-length variance, passive density, contraction avoidance, transition repetition, and document-level tricolon density, all as counts with thresholds. About forty “Figurative*” rules implement stop-slop’s false-agency idea programmatically (inanimate subjects with human verbs, one verb family per rule so each span produces one alert). A separate commit-message style (“This commit adds...,” “...ensuring consistency”) transfers directly to our multi-agent commit hygiene.

Four stock rules are destructive for our use and must be disabled or downgraded for LaTeX, and it is worth recording exactly why. **EmDashUsage** fires at error level on every `em` and `en` dash, so every numeric range fails. **DoubleHyphen** flags literal `-`, which in LaTeX source is

the correct en-dash syntax; it would flag 1990–2020 in every bibliography. **FormalRegister** bans, at error level, a token list including “utilize,” “facilitate,” and “implement,” and in the same list “minimize” relatives; an estimator that minimizes an objective cannot be discussed under that rule. **FormalTransitions** fights the existence of “moreover/furthermore” when the tell is clustering. All four are two lines each in `.vale.ini`, which is precisely the precedence mechanism as configuration.

The one real gap is LaTeX itself. Vale has no native `.tex` parser (verified in its format registry, which lists markdown, HTML, reST, AsciiDoc, and friends, but not TeX). The workable options are mapping `tex` to the markdown parser with block-ignore regexes for math environments, running Vale over detexed text at the cost of line fidelity, or hosting rules in a genuinely LaTeX-aware tool (Section 5.12). Piloting that choice is an explicit early task in WP2.

5.7 seyedehsanhadi/sloptrim: the deterministic scorer

Single-file Python detector (2,223 lines, standard library only) plus Claude Code hooks; Apache-2.0; 178 stars; active, 119 tests in CI on three platforms. Verdict: adopt.

sloptrim is the tool from Table 3. Its detector reads a file (twenty formats, including `.tex`, `.docx`, and `.ipynb`; prose is extracted, code is skipped), scores the first 256 KB against its pattern catalogue, and emits JSON: per-pattern counts with sample spans, structural metrics (sentence-length coefficient of variation, paragraph monotony, opener repetition, hedge stacking, synonym cycling, transition clusters, cadence zigzag), and a 0–100 score in five bands with a word-count-aware confidence label; a flag adds bootstrap confidence intervals. The hook layer wires the detector into Claude Code so every file saved through the editing tools is scored on save and flagged spans are surfaced to the agent; a documented blind spot is that files written by shell redirection bypass the hooks.

Two properties earn the adopt verdict beyond the score itself. The pattern accounting is honest: 71 documented patterns, 62 with detectors, and only 50 allowed to move the score, because the project measured the other 12 against human formal writing and found they mark register rather than authorship, demoting them to advice. That is our precedence principle, discovered independently by measurement, inside the tool. And the project publishes its own limits: a benchmark with ROC-AUC by arm (0.76–0.95), an explicit statement that it is not an authorship classifier, and an ethics file warning against consequential use.

Observed behavior on LaTeX, from our runs: equation environments and inline math pass through unmolested (the `-clean` mode touches characters only, never wording), which is the property we most needed. Two artifacts to correct for: pattern words inside `\text{...}` are scanned as prose, and on raw `.tex` the opener-repetition metric can count command sequences as sentence openers (our run flagged “`\item \textbf`” repetition in a list). Scoring detexed text alongside raw source removes both. The tool slots into **humanvoice** as the on-save scorer and the first layer of **hv rhythm**, with the calibration of Table 3 rerun whenever its pattern set updates.

5.8 AIScientists-Dev/academic-humanizer: the missing academic layer

One 261-line skill plus examples; MIT; 1.1k stars; last push July 2026. Verdict: adapt, by merging its content into our policy layer.

This is the one evaluated project written for our genre, and it opens with the sentence our whole stack is organized around: “academic writing already has a correct human voice: neutral, precise, third-person plural (‘we’), every claim tied to its evidence.” No personality injection, no casualization, explicitly “not for evading AI-use disclosure.” Its value is three layers our installed humanizer does not have.

Layer 2 is an academic-specific tell catalogue, and its examples are chosen well enough to quote. Over-claiming verbs: “We prove that our method significantly outperforms all prior approaches” becomes “Our method improves held-out accuracy by 4–7 points over the strongest prior approach (Table 3); the gain is significant at $p < 0.01$ by a paired test.” Significance hype (“paves the way,” “sheds light on”), empty intensifiers (“extensive experiments on a wide range of datasets” becomes “we evaluate on three datasets, named”), novelty padding (“to the best of our knowledge”), formulaic openers (“In recent years, X has attracted increasing attention”), connective *clustering* (the correct formulation, against consecutive-sentence connectives rather than the words themselves), citation dumping, contribution-list clichés, and, notably, clause-stacked sentences past thirty words, with the instruction to split. Every one of these appears in our agents’ drafts; none is in Wikipedia’s catalogue.

Layer 3 is a preserve-list that functions as a built-in precedence header: keep evidence-tied hedging (“suggests,” “is consistent with”), and it names the failure mode of general humanizers, “fixing” a calibrated hedge into an over-claim; keep passives where the actor is irrelevant; keep “we”; keep definitions, notation, and equations verbatim; never alter a number or citation key. Layer 4, claim-evidence discipline, checks every empirical claim for a backing number, figure, or citation and downgrades any verb stronger than its evidence, preferring attributed ranges to bare averages. This is our non-evasion policy meeting our meaning-ledger skill, written independently. Layer 6 is a funding-proposal mode with a different register contract (vision plus feasibility, aims that survive their siblings’ failure), useful whenever the group writes grants.

Weaknesses, for the record: a flat em-dash removal rule (though its own examples use LaTeX en dashes correctly); no diagnostics of any kind, so everything rides on the editing model’s compliance; machine-learning flavored examples rather than economics; and a single-file young project. None of that argues for running it as a second skill beside humanizer, which would create rule-conflict ambiguity; all of it argues for merging Layers 2–4 into our policy stack as the scholarly layer, with examples transposed to econometrics. That merge is a WP1 task.

5.9 conorbronsdon/avoid-ai-writing: three ideas worth taking

An 813-line skill with 62 pattern categories plus measurement scripts; MIT; 3.2k stars; active. Verdict: mine for patterns.

avoid-ai-writing is the most engineering-minded of the general skills: it ships false-positive measurement against public human/AI corpora and a CI check on its own pattern count. Three of its ideas transfer.

First, the tier system. Its 112-entry word table splits “always replace” into two bands with different *meanings*: 1A are frequency markers (“delve,” “tapestry”) whose clustering is evidence about how a passage was produced; 1B are clarity edits (“utilize,” “commence”) that improve any prose but, measured against 257 paragraphs of verified pre-2023 human writing, fire on ordinary formal prose at a meaningful rate, so the detector “weights them like Tier 2 and excludes them from the dense-AI-vocabulary signal so a wordiness fix can never push a document toward an AI classification.” Same edit, different evidentiary weight. Our diagnostics

should copy this distinction exactly.

Second, the injection boundary. In its file-editing mode, text under audit that addresses its editor (“ignore the rules above,” “don’t flag this section”) is flagged rather than obeyed, with instructions accepted only from the invoking writer. Any multi-agent editing pipeline needs this rule stated, and ours will state it.

Third, flag-don’t-fix zones: quotes, code, tables, and text attributed to others are reported but never rewritten, with the nice observation that a tell inside a table cell is not worth the risk to the data the table carries. Combined with a genuinely careful treatment of weak signals (curly quotes and flawless typography are “corroborating, never conclusive,” and when editing a human’s casual text, preserve their typos, since smoothing erases the fingerprint that marks the text as theirs), the skill shows what falsification-aware pattern work looks like. We still do not install it: its context profiles run from LinkedIn to investor emails with no academic register, and its dash policy (“Target: zero”) is wrong for us. The three ideas come; the skill stays.

5.10 NulightJens/humanizer-stack: the structural thesis

Two chained skills (surface pass, structural pass) plus two small deterministic scanners; MIT; 226 stars. Verdict: mine for patterns.

humanizer-stack’s premise is the StoryScope result reported in Section 4.1: word-level tells are the decaying, easily-fixed half of the fingerprint, and discourse structure is the durable half, so run a surface pass (its first skill is a humanizer variant) and then a structural pass. The structural skill audits six discourse habits with human-versus-AI base rates quoted from the study: themes stated and moralized rather than left to land, single-track tidy arcs with everything resolved, no digressions, unbroken linearity, and, its deepest point, *shape convergence*: all five models it examined occupy one tight region of structural space while human pieces are dispersed. Its operating instruction follows: pick one or two structural interventions per piece and vary them across pieces, because a fleet of documents fixed the same way converges on a new detectable cluster.

For our genre the six audits mostly do not transfer; a methods section is supposed to be tidy and linear, and IMRaD structure is a feature. Two things transfer anyway. The anti-convergence warning is directly relevant to *humanvoice*, which will edit every manuscript the group produces with one toolkit; measuring cross-document structural similarity is the guard, and it enters the design as part of *hv drift*. And the two scanners (`copy_scan.py`, `structural_scan.py`) are small clean reference implementations of deterministic tell-counting, worth reading if not importing. The skills themselves stay out: their genre calibration is narrative and marketing, and their base rates do not describe scholarly prose.

5.11 Corpus instruments: slop-forensics and fast-detect-gpt

sam-paech/slop-forensics: Python toolkit for deriving over-representation lists from model output; MIT; 369 stars. Verdict: adapt, corpus-level. baoguangsheng/fast-detect-gpt: research code for the ICLR 2024 detector; MIT; 424 stars. Verdict: adapt, aggregate-only, under guardrails.

slop-forensics inverts the stale-ban-list problem. Instead of consuming someone’s 2024 word list, it takes a corpus of model outputs, compares word, bigram, and trigram frequencies against human baselines (via `wordfreq` and NLTK), computes vocabulary-complexity and slop-index metrics, and emits the over-representation list for *your* models on *your* tasks; it can even cluster models phylogenetically by slop profile. Pointed at our own agents’ drafts with a pre-2022

economics baseline instead of general English (its stock baseline would flag “heterogeneous” and “endogeneity” as anomalies), it becomes a self-updating tell list, which matters because the tells decay: “delve” is already fading as vendors patch it, and yesterday’s list quietly loses power. This is the Kobak method of equations (1)–(2) as maintained software, minus the counterfactual projection, which we add ourselves in `hv drift`.

fast-detect-gpt’s repository contains clean reference implementations of the curvature statistic of equation (3), including a local inference script that runs with small surrogate models on CPU. Under Section 4.3’s findings, its only legitimate role here is as an aggregate drift instrument: a quarterly curvature distribution over detexed manuscript prose, watched for trend, never reported per-document, never attached to a name, and interpreted with the standing caveat that polished scholarly register inflates false positives by construction. If that discipline ever feels burdensome, the correct response is to drop the instrument, not the discipline. (Binoculars, the other zero-shot detector we cloned for comparison, is stale since 2024 and needs a heavier model pair; fast-detect-gpt supersedes it for local use.)

5.12 The LaTeX question: TeXtidote and textlint

TeXtidote: LaTeX-aware grammar and style checker wrapping LanguageTool; GPL-3; 1.1k stars; active. *textlint*: pluggable prose linter with a LaTeX AST plugin; MIT; 3.2k stars core, plugin small; active. Verdict for both: pilot.

Section 5.6 left one problem open: none of the AI-tell rule hosts parses LaTeX natively. Two tools do. TeXtidote reads `.tex` directly, skips math environments, maps every finding back to source lines, and wraps LanguageTool, whose custom-rule mechanism (XML regex rules) could host simple tells with real LaTeX awareness; GPL-3 is fine for internal use. textlint is the opposite trade: a first-class custom-rule ecosystem in JavaScript with a true LaTeX parser plugin (`textlint-plugin-latex2e`), but no existing AI-tell rule pack, so everything we want would be written by us. The pragmatic WP2 sequence is: start with Vale over detexed text plus sloptrim on raw source (both work today), and pilot TeXtidote as the line-accurate LaTeX layer; move rules into textlint only if the pilot shows the detex mapping loses too much.

5.13 proselint, and the rest

proselint (BSD-3, 4.6k stars) is the classic usage linter, 87 checks distilled from Garner and the style canon: archaisms, clichés, redundancy, weasel words. It predates LLMs, contains no AI-tell content, and, decisively, has no custom-rule mechanism, so extending it means forking it; since a Vale-compatible re-implementation exists, anything we want from it arrives through the Vale configuration for free. Skip, and inherit via Vale if ever wanted. The remaining survey finds were language ports of humanizer (a 15.8k-star Chinese localization among them), smaller Vale slop styles superseded by vale-ai-tells, and academic humanizer variants less developed than the one reviewed above; none changes a verdict.

5.14 Verdicts, and what adoption means

Put together: humanizer stays as the installed general diagnostic; academic-humanizer’s content becomes our scholarly layer; sloptrim and a curated Vale configuration become the measurement floor; slop-forensics and the mixture estimator keep the measurements current; three skills contribute patterns without being installed; two LaTeX-aware hosts get a pilot; and three

Project	Verdict	What the verdict means in practice
blader/humanizer	adopted	keep vendored copy; precedence header stays; enforce dash override in config
tbhb/vale-ai-tells + Vale	adopt	CI linter; disable four rules for <code>.tex</code> ; enable structural metrics; take commit rules
seyedehsanhadi/sloptrim	adopt	on-save scorer and rhythm layer; recalibrate per Table 3 on updates
AIScientists-Dev/academic-humanizer	adapt	merge Layers 2–4 into our policy as the scholarly layer, examples transposed to economics
sam-paech/slop-forensics	adapt	derive our own tell lists against a pre-2022 economics baseline
baoguangsheng/fast-detect-gpt	adapt	aggregate drift metric only; never per-document
conorbronsdon/avoid-ai-writing	mine	take 1A/1B weighting, injection boundary, flag-don’t-fix zones
hardikpandya/stop-slop	mine	take false agency (with econ carve-out), negative listing, Wh-openers
NulightJens/humanizer-stack	mine	take the anti-convergence diagnostic; skills stay out
sylvainhalle/textidote	pilot	candidate line-accurate LaTeX layer
textlint + latex2e	pilot	fallback host if detex mapping proves lossy
amperser/proselint	skip	no custom rules; content reachable via Vale
leonxlnx/taste-skill	skip	frontend pack; one sociological lesson retained
itallstartedwithaidea/writing-agent	skip	evasion-framed; weak checks; nothing to mine

Table 4: Verdicts over the evaluated open-source field, with the concrete action each verdict implies.

projects are set aside for cause. Nothing in the field does what Sections 2 and 3 showed we need most, reader-based acceptance testing and behavioral evaluation of writing doctrine, which is why the proposal builds those two things and mostly configures the rest.

6 What the evidence implies for the design

Everything above now compresses without loss. The failure record says the defects are layered and the deep layers decide acceptance; the literature says the shallow layers are measurable and the deep ones are only reachable by editing and by readers; the tool field supplies the shallow measurements nearly for free and supplies nothing for the deep layers. So *humanvoice* is a two-layer design. A *measurement layer* of deterministic diagnostics runs constantly, reports numbers with locations, and gates nothing. An *acceptance layer* of context-starved reader proxies and, finally, the human reader gates everything. Between them sits an editing workflow that repairs in layer order under protected baselines.

Table 5 states the requirements this survey commits the design to, each with its evidence. They are numbered for reference from the work packages; none is aspirational, in the sense that every one of them traces to a failure we paid for or a result we verified.

One requirement deserves a paragraph rather than a row, because it is the one a tooling project most easily forgets. The acceptance metric for *humanvoice* itself is the number of human-feedback rounds a document needs before its reader accepts it. The ZLB rescue took five. SMEwallet’s units took one to three each. The BGS volume never arrived. Every work

#	Requirement	Evidence
R1	Two documents: internal record and human manuscript, tied by an identifier map; ledger vocabulary never crosses	§2.2.1, §2.4.1
R2	Precedence as configuration: every pattern source enters as a subordinate diagnostic, with per-format overrides in config, not prose headers	§3.4, §5.6
R3	Diagnostics report counts with locations and never auto-fix; repairs are made by an editing agent and verified by re-measurement	§2.6(9), §5.4
R4	Repair follows the layer order: register, mechanical tells, defensive prose, substance	§2.4.1
R5	Protected baselines on every pass: equation-body identity, label and citation censuses, explain-every-removal	§2.6(10), §2.3
R6	All corpus statistics are referenced to pre-2022 economics text; excess frequency per eqs. (1)–(2); prevalence per eq. (4)	§4.1, §4.3
R7	The editor is span-diagnose-then-edit with categories; expected span precision near 0.5; flags adjudicated, not auto-applied	§4.4
R8	Model judging is pairwise, order-randomized, length-controlled, cross-family, one dimension per call, validated against recorded human verdicts	§4.5
R9	Behavioral evaluation of every skill against the corpus before it is called an improvement	§3.8
R10	Thin process: one plan and one result note per work package; no mission-control state; ceremony is a register contaminant	§2.2.4
R11	No per-document authorship scores, ever; population statistics only, firewalled from attribution	§4.3
R12	Cross-document convergence is monitored, so the toolkit does not become the new template	§2.6(4), §5.10

Table 5: Design requirements with their evidence. Section references point to where this survey establishes each one.

package below either reduces that number or exists to measure whether something else did; any proposed activity that does neither is out of scope by construction, which is the practical meaning of R10.

7 The proposal

7.1 Shape of the repository

humanvoice consolidates into five trees, all under the existing repository. **policies/** becomes the single home of the writing doctrine: the prose policy’s reader-facing sections, the two claudex skills, MathDevMCP’s repair skill, the installed humanizer list, and a new scholarly layer merged from academic-humanizer’s Layers 2–4 with examples transposed to economics (R2 makes claudex the distributor, no longer a source of truth). **diagnostics/** is the measurement layer, one CLI named **hv**. **corpus/** holds the evaluation assets. **harness/** holds reader-proxy templates and the behavioral tests. **vendor/** stays as the pinned external tools with their manifest.

7.2 The hv commands, specified

Each command below lists what it consumes, what it emits, what it is built from, and how we will know it works. All of them obey R3: numbers out, no edits in.

hv leak consumes rendered text (pdftotext output) plus the manuscript source; emits counts and locations of internal-register leakage: file paths, dates-as-scoping, identifiers, and a project-configurable banned-vocabulary map seeded from the ZLB style contract’s ban list and the BGS audit’s administrative-register examples. Built as a small matcher over per-project YAML config. Acceptance: on the ZLB pre-rescue markdown it must fire heavily; on the accepted final PDF text it must report zero, reproducing the recorded leak-sweep result.

hv rhythm consumes source and detexed text; emits the mechanical-tell profile: dash counts, sentence-opener and paragraph-opener distributions, section-shape clustering, transition-phrase counts, sentence-length mean and coefficient of variation, paragraph monotony, tricolon density. Built by orchestrating sloptrim and the curated vale-ai-tells experimental metrics, with the four destructive rules disabled for `.tex` (R2) and the detex path handling the artifacts observed in Section 5.7. Acceptance is historical replay: on the preserved ZLB pass-two snapshot it must reproduce the recorded numbers, 133 em dashes and 20 of 55 sections opening with “The”; on the final manuscript, 0 and at most 4 per opener word; on the three calibration documents, sloptrim’s 45/25/10 within tolerance.

hv defensive consumes detexed text; emits candidate sentences with surrounding context for the defensive-register classes: negative definitions without mechanisms, prohibitions without consequences, the moralizing tics, unraised-objection answers (humanizer patterns 34–35, the DefensiveHedges shapes, and the BGS defensive-qualification examples as seeds). Each candidate is emitted with the editorial question attached, “does this sentence give the reader a reason, or only protect the author?”, for an editing model or human to adjudicate (R7: flags, not fixes). Acceptance: on the preserved ZLB pass-three text it must recover a majority of the 65 sentences the manual diagnostic found, and its false-positive rate on the accepted final text must be low enough that an editor reviews its output without resentment; both measured, with the corpus, in WP3.

hv diff consumes two versions of a manuscript; emits the protected-baseline report: equation bodies byte-compared, label, citation, and reference censuses, and an explain-every-removal list of substantive deletions (R5). Built on LaTeX-aware extraction, piloted per Section 5.12. Acceptance: replaying the five ZLB passes must show 109 of 109 equations identical at every step and flag the deliberate reference additions of the final pass and nothing else.

hv gate consumes the manuscript; emits the compile gate (N-pass build, unresolved references, overfull-box budget) and rendered page images for the title, a math-dense page, and every table, institutionalizing rendered-page inspection. Built from the repair skill’s two scripts (Section 3.5). Acceptance: reproduces the recorded ZLB build-gate outcomes.

hv drift consumes a corpus of our manuscripts plus the references; emits quarterly population statistics: excess-vocabulary profiles against the pre-2022 economics baseline (eqs. (1)–(2), via

adapted slop-forensics), the mixture-estimated machine-typical fraction $\hat{\alpha}$ of eq. (4) with Q estimated from our own agents’ raw drafts, cross-document structural similarity (R12), and optionally an aggregate curvature distribution. It has no per-document mode, by construction (R11): the command refuses single-file input. Acceptance: on a synthetic mixture of known composition, $\hat{\alpha}$ within the validation error reported by Liang et al. [2024a].

7.3 Work packages

WP1: consolidation. Move the doctrine into `policies/`, merge the scholarly layer, convert the humanizer precedence header into diagnostic configuration, add Claude-native skill installation beside the Codex path, and leave claudecodex as pure distribution. Deliverable: one install command produces both agent environments from the `humanvoice` source of truth; the existing phrase-lock tests pass against the new layout. Effort: small; risk: low; unblocked today.

WP2: the measurement layer. Build the six commands above, in the order `leak`, `rhythm`, `diff`, `gate`, `defensive`, `drift`, because the first four are configuration and replay while the last two need corpus work. Includes the LaTeX-host pilot (Vale-on-detex versus TeXtidote). The acceptance tests are the historical replays listed per command; a tool that cannot reproduce the recorded history does not ship. Unblocked today; the replay fixtures (ZLB snapshots) already exist in the BayesFilter repository.

WP3: corpus and behavioral evidence. Assemble `corpus/`: the 815 BGS repair pairs with their taxonomy labels, the ZLB pass snapshots with their measured counts and final acceptance, the SMEwallet draft-10/draft-11 units with recorded PI verdicts, and a pre-2022 economics reference corpus. Then run the experiment the stack has never had (R9): matched rewrites of corpus units with and without each skill, scored by `hv`, judged pairwise under the R8 protocol, the judge itself validated against the recorded human verdicts. Deliverable: a scorecard per skill stating what it measurably changes, plus the convergence baseline for R12. Needs WP2’s scorers.

WP4: the editorial workflow. One skill codifying the pipeline the record validated: two-document setup with identifier map (R1), a drafting contract in the ZLB style-contract form, repair in layer order (R4) with `hv` measurements before and after each pass, a single voice editor after any parallel drafting, per-agent drop reports, protected baselines via `hv diff` (R5), then low-context reader-proxy review descended from the AIpostdoc reader-pull personas with forced choices and the injection boundary, and the human read as the only promotion (R8’s proxy explicitly cannot promote). Mostly consolidation of existing text; the work is deleting the ceremony from it (R10).

WP5: the pilot. Apply the full pipeline to one live document, the next BayesFilter companion note or one CIP monograph part, with the ZLB rescue as benchmark: five feedback rounds to acceptance. The pilot’s single question is whether the toolkit reduces that count. Its result note, one page, states the count, what consumed the rounds, and which tools earned their place. Needs WP1, WP2, WP4; WP3 can run concurrently.

WP1 and WP2 start immediately and independently. Per R10, each package carries one plan and one result note; a package that begins generating review packets, milestone registries, or state files is failing by definition, and the correct repair is deletion, not documentation.

7.4 Risks, stated with their controls

Four risks have known shapes, because we have watched each one happen. The toolkit becomes the new template and the fleet converges on a detectable house style: R12’s convergence monitoring exists for this, and R3 (report, never auto-fix) keeps a human or editing agent’s judgment in every repair. The gates ossify and get gamed, the BGS checkpoint story: diagnostics here gate nothing, so there is nothing to game except the human read, which cannot be gamed, only failed. The project spends itself on instrumentation, the BGS process story: the WP budget is the one-page plan-and-result cap, and the pilot metric is the only scoreboard. And the drift instruments get misused for attribution: R11 firewalls them, the drift command refuses per-document input, and if the discipline ever slips, the instrument goes, not the firewall.

What this proposal does not claim, so that nobody reads more into it than the evidence holds: no tool here, ours or adopted, has yet been shown to reduce human-feedback rounds; that is exactly what WP3 and WP5 measure. The literature’s load-bearing results were verified at method level, the supporting ones at abstract level, and StoryScope remains secondhand. The sloptrim accuracy figures are its own benchmark. And the reader-proxy reviews, however well they predicted one human rejection, are proxies: the record contains one case of proxy-predicts-human, not a validation study, which is another thing WP3’s corpus makes testable.

8 Conclusion

The record, the literature, and the tool field agree on more than we had any right to expect. Our five projects learned by expensive trial and error that writing failures arrive in layers and that review machinery cannot write; the research literature explains why the models write this way and shows that targeted editing, population-level measurement, and disciplined judging work where detection and word bans do not; and the open-source field has independently converged on the same division of labor, with pattern lists for editors, deterministic counters for pipelines, and honest warnings against using any of it as an authorship verdict.

What no source establishes, including this one, is that the assembled toolkit reduces the number of human-feedback rounds a document needs. That is the only outcome worth the project’s budget, it is measurable, and work packages three and five exist to measure it. The survey’s own history sets the benchmark and the caution in one stroke: its first version passed every mechanical check we have and failed its only reader. The reader remains the gate. Everything this proposal builds is machinery for wasting less of the reader’s time.

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